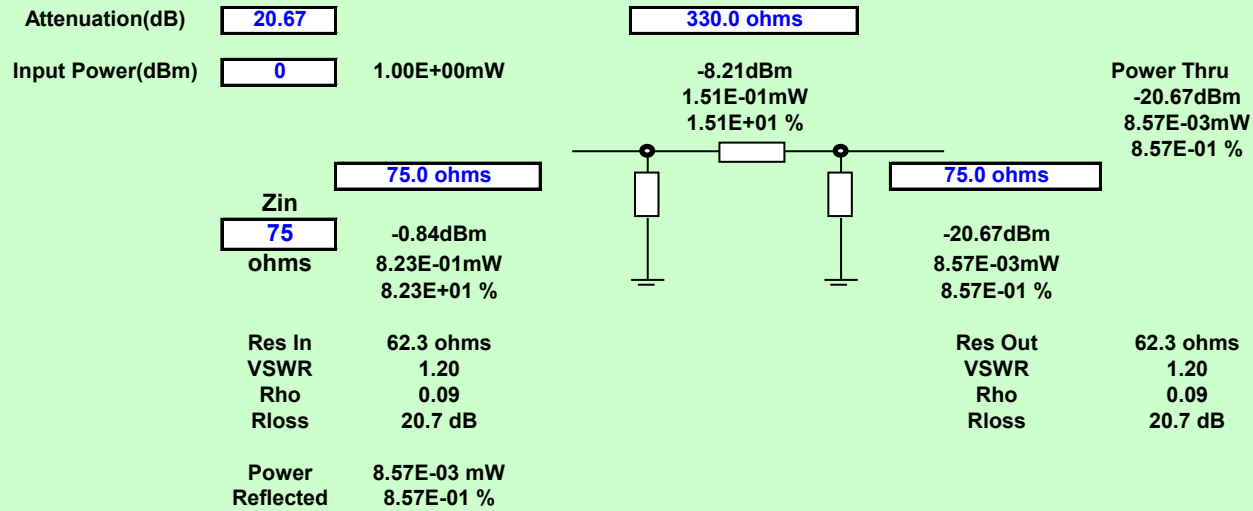


PI PAD Attenuator (ver 3.2)

April 2009

<http://www.rfshop.co.uk>

T PAD Attenuator (ver 3.1)

Attenuation(dB) 7.56

Input Power(dBm) **15** 3.16E+01mW

Zin
50
ohms

Res In	50.00
VSWR	1.00
Rho	0.00
Rloss	>60 dB
Power	0.00E+00 mW
Reflected	0.00E+00 %

The diagram shows a power divider circuit. An input signal 'W' enters from the left. The circuit splits into two parallel branches. The top branch contains a 20.5 ohm resistor, with output power 3.57dBm (2.27E+00mW, 7.2%). The bottom branch contains a 20.5 ohm resistor in series with a 50.8 ohm load, with output power 11.12dBm (1.29E+01mW, 41.0%). The total output power is 7.44dBm (5.55E+00mW, 17.6%). The load is labeled 'Zout 50 ohms'.

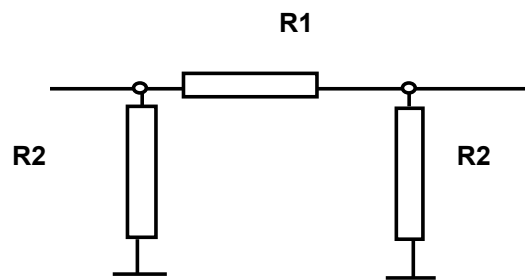
Component	Value	Power (dBm)	Power (mW)	Efficiency (%)
Input	-	-	-	-
Top Resistor (20.5 ohms)	20.5 ohms	3.57dBm	2.27E+00mW	7.2%
Bottom Resistor (20.5 ohms)	20.5 ohms	11.12dBm	1.29E+01mW	41.0%
Bottom Load (50.8 ohms)	50.8 ohms	10.35dBm	1.09E+01mW	34.3%
Total Output	-	7.44dBm	5.55E+00mW	17.6%

Res Out	50.00
VSWR	1.00
Rho	0.00
Rloss	>60 dB

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PI PADS USING STANDARD VALUE RESISTORS

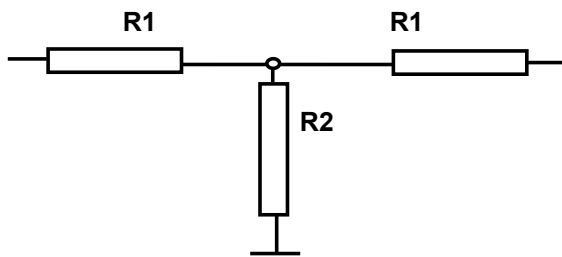


dB loss	RL /dB (Ideality)	R1	R2
0.5	>50	3R3	1k8
1.0	46	5R6	820R
2.2	42	12R	390R
3.1	42	18R	270R
3.9	38	22R	220R
5.2	36	33R	180R
6.1	43	39R	150R
7.5	38	47R	120R
8.1	35	56R	120R
9.6	49	68R	100R
12.5	47	100R	82R

System Impedance =50ohms

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T PADS USING STANDARD VALUE RESISTORS



dB loss	RL /dB (Ideality)	R1	R2
0.4	>50	1R0	1k2
1.0	38	3R3	470R
2.2	34	5R6	180R
2.9	>50	8R2	150R
3.5	>50	10R	120R
4.2	>50	12R	100R
5.2	38	15R	82R
6.2	34	18R	68R
6.8	38	18R	56R
8.1	47	22R	47R
10.5	>50	27R	33R
11.4	30	27R	27R
13.6	48	33R	22R

System Impedance =50ohms

<http://www.rfshop.co.uk>

Notes:

Metal Film Leaded resistors are usually OK up to 100 MHz

1206 SM Resistors are good up to about 1GHz.

0805 SM Resistors give good results up to 1.5GHz

Above this 0402 can be used but have low power rating.

1206 and 0805 can still be used up to a few GHz but allowance needs to be made for parasitic inductances and capacitances.

This limits the frequency range.

High value pads can be difficult at high frequencies.

Often it is better to use 2 x 10dB (one PI & one T) rather than one 20dB pad.